



Draft genome of *Serratia* sp. R1 gives an insight into the antibiotic resistant genes against multiple antibiotics

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Abstract

Background *Serratia* is a pathogenic bacterium, commonly associated with neonatal intensive care units, and harbors antibiotic-resistant genes against multiple antibiotics e.g., resistance against penams, aminoglycosides, tetracyclines, cephalosporins, and macrolides. In the long-term contaminated habitat, the bacterial communities carry both antibiotic and metal resistance genes. This draft genome sequencing aimed to explore the alarming level of ARGs in the environment, additionally heavy metal-resistant genes were also explored in the draft genome.

Methods Whole-genome sequencing was used to investigate ARGs in *Serratia* sp. R1. The bacteria were sequenced using Illumina Nova seq sequencer and subjected to genome annotation. The bacterial genome was explored for antibiotic- and metal-resistant genes.

Results Sequencing resulted in 8.4 Mb genome and a total of 4411 functional genes were characterized in the draft genome. Genes resistant to Beta-lactams, cephalosporins, macrolides, fluoroquinolones, and tetracycline are present in the draft genome. Multiple metal-resistant genes are also present in the sequenced genome.

Conclusion The genes and proteins providing heavy metal and antibiotic resistance may be used in the bioremediation of environmental antibiotic residues to prevent the spread of antibiotic resistance. The current study can help us to adopt suitable mitigation measures against the multidrug-resistant *Serratia*.

Keywords Antibiotic-resistant gene · Antibiotic-resistant bacteria · *Serratia* · Sequencing

Abbreviations

ARGs Antibiotic-resistant genes
ARB Antibiotic-resistant bacteria
CARD Comprehensive antibiotic-resistant database

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Introduction

Serratia, a Gram-negative, motile, encapsulated, pigment secreted, anaerobic, and facultative aerobic, non-sporulating bacteria, belongs to the family Enterobacteriaceae [1]. In addition to both the environmental matrices, soil, and water, it is also found in plants, food, garbage, moisture places [2]. *Serratia* was not considered pathogenic until the 1970s, but it came as an emergent pathogen and was also included in the list of world health organizations for antibiotic resistance. It is an opportunistic pathogen and is supposed to do nosocomial infections in humans [3]. *Serratia* is now resistant against multiple antibiotics, starting from the first, Penicillin to third and fourth generation antibiotics. Due to its pathogenicity and resistance against multiple antibiotics, *Serratia* causes serious infections in humans [4]. *Serratia* is an emergent pathogen and has a role in hospital-acquired infections [5]. The hospital waste which finds its way to the aquatic environment via the sewage treatment plants is the reservoir of the pathogenic bacteria. *Serratia* is known to harbor multiple antibiotic-resistant genes (ARGs) and due

to this, it becomes resistant against multiple antibiotics, e.g., β -lactams, aminoglycosides, fluoroquinolones. There are many species of *Serratia*, but *Serratia marcescens* is very well explored by researchers [6].

Antimicrobial resistance is a global problem and poses a threat to human and animals health. Due to antibiotic resistance, the bacterial infection becomes incurable in some cases. Moreover, the reduction in modern drug discovery has limited the therapeutic options. *Serratia* secretes the red pigment, prodigiosin [7]. *Serratia* has been investigated by researchers for the virulence factors of the prodigiosin.

In this study, the isolated strain in the previous studies of [2], was sequenced on the Illumina-Hiseq platform and gained an insight into the antimicrobial-resistant genes and heavy metal resistant genes in the organism. The current study also explores the genomic details of *Serratia marcescens* which may lead to the development of strategies to prevent the spread of antibiotic resistance of this infection-causing bacteria, e.g., exploring the number of antibiotic resistance genes in *Serratia marcescens* may help to check the pattern of variation of ARGs in other species of *Serratia*. The tracking of newly acquired ARGs may explore the pathways of acquiring antibiotic resistance and hence, may lead to development of the strategies to overcome the antibiotic resistance spread in the environment and bacterial community. This approach may promise a safe and healthier environment.

Materials and methods

Isolation and identification

The bacteria used in this study was isolated and identified as described by Kumar et al. [2]. The water samples were collected from the contaminated Yamuna River, Delhi at the Wazirabad stretch (28°40'5.53" N, 77°15'0.35" E).

Whole genome sequencing and annotation of *Serratia* sp. R1

Serratia sp. strain R1 was cultured on Luria–Bertani broth media (yeast extract (5 g/L), tryptone (10 g/L), sodium chloride (5 g/L); and pH 7.2–7.4) at 28°C. DNAasy extraction kit (Qiagen-United States) was used to isolate the bacterial genomic DNA. The overnight grown culture was subjected to DNA isolation using a DNAasy extraction kit (Qiagen-United States) according to manufacturer protocol. The quantification was performed using Qubit Fluorometer. The Whole-genome sequencing library was prepared with the use of the KAPA HyperPlus kit. ~100 ng of DNA was taken as input and subjected to enzymatic fragmentation. The fragments were end-repaired, followed by A tailing.

Ligation was done and post ligation the samples were cleaned using KAPA clean-up beads. The adaptor-ligated library was amplified using PCR. The final libraries were then cleaned using KAPA clean-up beads. The quality of the library was checked using Qubit Fluorometer. Sequencing was done on Hiseq-X sequencing platform to generate paired reads of 151 bp read length by Illumina's Sequencing by Synthesis technology. Sequencing was done on Illumina nova sequencer 6000. Quality checks were done using the fastp tool (version 0.20.0) to get good quality raw data with a Phred score above 30. The sequence assembly to produce contigs and scaffolds from reads is done using Unicycler. The nearest phylogenetic neighbor was predicted using a 16S rRNA gene sequence using EzBioCloud (EzTaxon: a web-based tool for the identification of prokaryotes based on 16S rRNA gene sequences [8] and NCBI BLAST.

ARGs detection in the *Serratia* sp. R1

To identify the ARGs in the assembled genome of *Serratia* sp. R1, a Comprehensive antibiotic-resistant database (CARD) [9]. CARD 3.1.0 was used where the Open Reading Frame (ORF) prediction is done using Prodigal, homolog detection was done using DIAMOND, and Strict hits were used to detect a significance based on CARD curated bitscore cut-offs while assessing the resistance profiling. RGI at CARD site (<https://card.mcmaster.ca/analyze/rgi>), was used for mapping of resistome featuring homology.

Metal resistance genes exploration in the draft genome

BactMet2 database was used to identify the metal resistant genes in assembled *Serratia* genome. blastp was done against the resistance protein database (BacMet2) to fetch the metal resistance genes and antibacterial biocide genes.

Identification of secondary metabolite-biosynthetic gene clusters (BGCs)

AntiSMASH (antibiotics and Secondary Metabolites Analysis Shell) was used to identify and analyze the BGCs in the genomes of fungi and bacteria. In our study, the draft genome of *Serratia* sp. strain R1 was subjected to antiSMASH 5.0 using default parameters.

Results

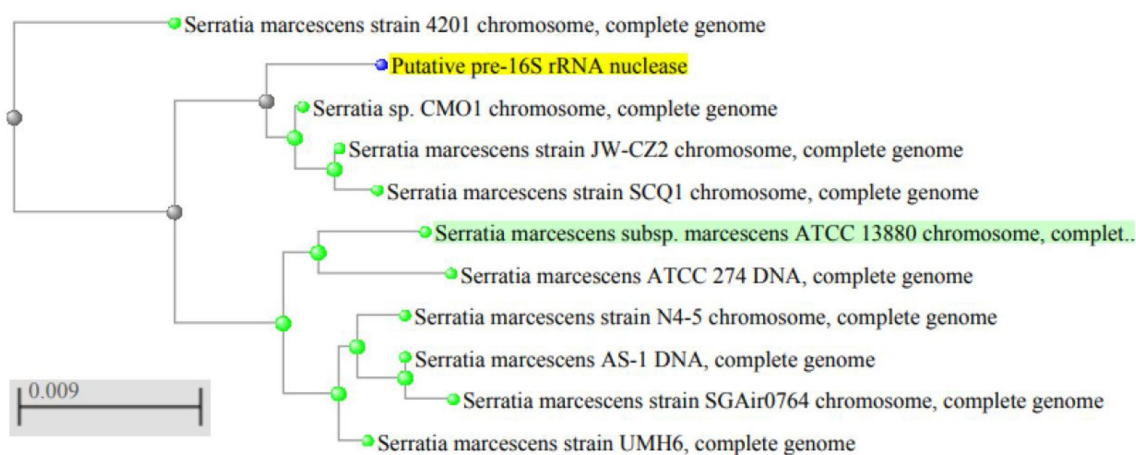
Draft genome, assembly, and annotation

During our previous study [2], the strain R1 was isolated from the river Yamuna and was identified by 16S rRNA

gene sequence as *Serratia* sp. R1 and can degrade up to 85% of penicillin G in 14 days. To further get an insight into the bioremediation capabilities of the *Serratia* sp. strain R1 whole genome sequence was employed and the total reads obtained were 8,320,992 and the read length was 151 bp. The complete genome assembly was done as de novo genome assemblies using short DNA sequences. The raw reads were generated and their quality was checked as quality filtering and adapter trimming. The work was done by fastp, FASTQ preprocessor, version 0.20.0 [10]. The total quality reads after quality filtering were 7,189,916. All the trimmed sequences were subjected to primary assembly, and the 16S rRNA genes were predicted. The predicted species using EzBioCloud was *Serratia rubidaea* strain DSM 4480 with a percentage identity of 99.86%. The assembled draft genome was used for the prediction of the total genes available. The final gene prediction led to protein function annotation. The primary assembly of high-quality filtered reads was done by SPAdes (version 3.14.1) for the long contiguous sequence. The primarily assembled genome produced 456 long sequences containing 45,581,169 bases with 60.51% GC content, and the assembly was used for the prediction of 16 s rRNA gene sequence (RNAmmer, version 0.9.24). Supplementary Fig. 1 shows the draft genome circular map of *Serratia*. Figure 1 shows the phylogenetic tree of R1, representing the *Serratia marcescens*. The assembled draft genome was further used for the gene predictions and annotations, using the RAST annotation web server, and obtained a high-quality genome annotation.

Functional annotation of the draft genome

A total of 4492 genes were predicted from the final assembled and annotated draft genome, out of which 4411 genes were the protein-coding genes, suggesting that the bacteria may have an active metabolic stage. The non-coding genes were found out to be 81. However, 4411 were the protein-coding genes but the characterized proteins were 3924, and this suggested that the remaining 487 genes were putative/hypothetical. The number of rRNA and tRNA were predicted as 6 and 75. The number of proteins with pathway annotation was 1497, while there were no genes predicted for, crisper array, crisper repeat, and crisper spacer. The repeat regions for the predicted genes were 26. The summary of the functional annotation of the draft genome is given in Supplementary Table 1. RAST annotation server showed 1497 proteins with annotation pathways, 26 repeat regions. Genome is uploaded in the RAST server, the automated server computes the similarities to all the genomes present in the SEED database. The annotation in the SEED database is based on manual curation. Subsystems are created, a set of proteins that performs a particular function within a genome. Subsystems are then examined for the functional variant in the available genomes in the SEED database. The genes related to stress response were also present. We identified multiple genes related to the biosynthesis of amino acids and cofactors, and protein metabolism. Genes involved in stress response were also present in *Serratia*. The genes related to the metabolism of aromatic compounds especially involved in the degradation of p-hydroxybenzoate and benzoate were also present in the bacterial strain. Supplementary Fig. 2 shows the subsystems overview obtained from genome annotation using the RAST tool kit.



Representing the Presence of *Serratia marcescens*

Fig. 1 The phylogenetic tree of R1, representing the *Serratia marcescens*

Presence of ARGs in the draft genome

Multi-drug resistance in bacteria has become a cause of serious concern. Bacteria are producers of antibiotics and can resist them for their survival. The strain was isolated from the contaminated river Yamuna. Sodhi et al. and Velpandian et al. [11–13] showed the occurrence of various antibiotics in the river. Therefore, the ARGs present in *Serratia* sp. R1 was cataloged to gain an insight into the environmental reservoir of such genes. Acquired AMR genes were identified using RGI, a tool for predicting resistome. The R1 assembly file was used as input in RGI based on that the wheel chart was drawn for AMR genes, drug class, etc. AMR genes showing resistance against penams, aminoglycosides, tetracyclines, cephalosporins, and macrolides are present in the *Serratia* sp. R1 and Table 1 show the antibiotic-resistant ontology (ARO) term for the *Serratia* sp. R1. ARGs along with their functions such as antibiotic efflux and beta-lactamase production. Figure 2 shows the depiction of antimicrobial-resistant genes of *Serratia marcescens*.

Heavy metals genes present in *Serratia* sp. R1

Various genes providing resistance against heavy metals such as copper, cadmium, etc. are shown to be present in the *Serratia* sp. R1 along with the multidrug efflux RND transporters. These genes are present in the *Serratia* sp. R1 draft genome indicates the bioremediation potential of the *Serratia* sp. R1. The bacteria were isolated from the site which is known to contain toxic heavy metals including copper, cadmium, chromium, nickel, and zinc [11, 12] so the bacteria which thrive in the heavy metal polluted environment

carry the genes which provide them resistance against multiple toxic metals, therefore these bacteria can be exploited for the clean-up of the polluted environments. The *Serratia* sp. R1 can be used for the bioremediation of heavy metals. Supplementary Fig. 3 and Table 2 show the heavy metal resistant genes in *Serratia* sp. R1.

BGCs presence in *Serratia* sp. R1 draft genome

The major BGC found was in Region 1 (Supplementary Table 2), which shows maximum similarity to prodigiosin, a heterocyclic secondary metabolite and belongs to the class of the tripyrrole compound and is synthesized by various bacterial genera, including *Serratia*. Prodigiosin has antimicrobial properties. It inhibits bacterial growth, virulence factor, biofilm production, and protease production, leading to antibiotic resistance. *S. marcescens* requires this pigment to compete against other bacterial species in their natural environment.

Discussion

Serratia is a well-known nosocomial pathogen and the main cause of many hospital-acquired infections. It is multi-drug resistant which makes the treatment of this pathogen impossible [14]. The significant concerns regarding the opportunistic nature of this pathogen are its ability to spread in the hospital environment, leading to severe outbreaks, and its potential to disseminate antibiotic resistance, combined with acquired antimicrobial genes and intrinsic mechanisms [15]. The bacterial strain

Table 1 The bacterial genes found resistant against a different class of antibiotics and the mechanism they employed for their resistance to ensure their survival under continuous exposure to heavy antibiotic load

S. no.	Antibiotic resistance ontology term	Antibiotic microbial resistance gene family (AMR)	Drug class	Mechanism of resistance
1	adeF	RND family antibiotic efflux pump	Tetracycline, Fluoroquinolone	Efflux of antibiotic
2	SRT-2	SRT β -lactamase	Cephalosporin	Inactivation of antibiotic
3	tet(41)	MFS antibiotic efflux pump	Tetracycline	Efflux of antibiotic
4	<i>Klebsiella pneumoniae</i> KpnF	MFS antibiotic efflux pump	Tetracycline, Cephalosporin, Aminoglycoside, Macrolide, Rifamycin, Peptide antibiotic	Efflux of antibiotic
5	rsmA	RND family antibiotic efflux pump	Phenicol, Diaminopyrimidine, Fluoroquinolone	Efflux of antibiotic
6	AAC(6')-Ic	AAC(6')	Aminoglycoside	Inactivation of antibiotic
7	<i>Klebsiella pneumoniae</i> KpnH	MFS antibiotic efflux pump	Cephalosporin, Aminoglycoside, Macrolide, Fluoroquinolone, Carbapenem, Peptide antibiotic	Efflux of antibiotic
8	CRP	RND family antibiotic efflux pump	Fluoroquinolone, Macrolide, Penam	Efflux of antibiotic
9	Haemophilus influenzae PBP3	Mutations in PBP confer resistance to β -lactams	Cephalosporin, cephamycin, penams	Efflux of antibiotic

Fig. 2 **A** The depiction of antimicrobial-resistant genes of *Serratia marcescens*, **B** Their antimicrobial-resistant gene family, **C** Drug class against which the antimicrobial-resistant genes were found and, **D** The antibiotic resistance mechanism of *Serratia marcescens* through which they are surviving under heavy antibiotic load in the environment

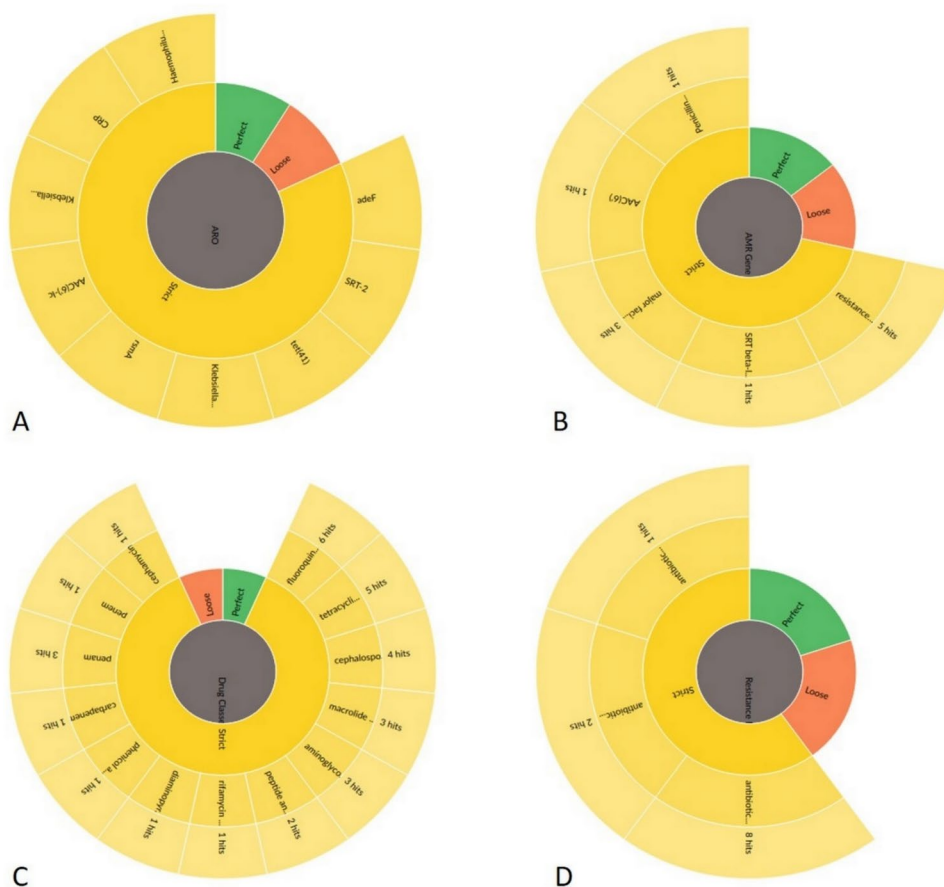


Table 2 Antibacterial biocide and metal resistant genes in *Serratia* sp. R1 using BactMet 2

S. no	Genes	Count
1	Copper-exporting P-type ATPase CopA [<i>Escherichia coli</i>] [copA]	9
2	DUF1471 domain-containing protein [<i>Serratia plymuthica</i>] [yhcN]	5
3	Iron ABC transporter ATP-binding protein FetA [<i>Klebsiella pneumoniae</i>] [fetA/ybbL]	5
4	Divalent metal cation transporter [<i>Salmonella enterica</i>] [mntH/yfeB]	4
5	DNA-binding response regulator [<i>Serratia marcescens</i>] [cpXR]	4
6	DNA-binding transcriptional regulator OxyR [<i>Serratia marcescens</i>] [oxyRkp]	4
7	DUF1471 domain-containing protein [<i>Leclercia adedecarboxylata</i>] [bhsA/ycfr/comC]	4
8	LPS assembly protein LptD [<i>Escherichia coli</i>] [ostA/lptD]	4
9	Multidrug ABC transporter permease/ATP-binding protein [<i>Serratia marcescens</i>] [smdB]	4
10	Multidrug efflux RND transporter permease subunit [<i>Serratia marcescens</i>] [sdeB]	4

was isolated from the River Yamuna, which has the prevalence of many antibiotics including the Beta-lactams and fluoroquinolones [16]. The *Serratia* sp. R1 was shown to degrade the antibiotic Penicillin G in Minimal media in 14 days [2]. The whole-genome sequencing of the *Serratia* sp. R1 shows that the strain is 4.6 Mb and is closest to *Serratia marcescens*. The genome annotations indicate many genes which can degrade xenobiotics and tolerate harsh and stressed conditions. The previous investigations

showed that this strain has bioremediation capabilities, and in-silico analysis using CARD and BactMet 2 also revealed ARGs and heavy metal resistance genes in its genome. Due to the emergence of multi-drug resistance among microbes, there is a huge demand for novel drug candidates. These results are in concordance with the studies by [17], who also showed that the active BGCs belonging Non-ribosomal peptide-polyketide. Prodigiosin was also the BGC and had antimicrobial properties.

Conclusion

The whole-genome sequence data of the bacterial strain has provided an insight into the ARGs present in the draft genome of *Serratia*, along with opening new vistas for the strain utilization for bioremediation of xenobiotics. The bioremediation using the bacterial strain is a sustainable and eco-friendly approach for the environmental clean-up of pollutants. The novel antimicrobial compounds can also be studied using *in-silico* analysis of the whole genome sequence of the strain and can further help to solve the problem of antimicrobial resistance.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11033-022-07290-8>.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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